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Automated Quality Assessment of Tri Dharma in Academic Accreditation Using Comparative Performance Study of Qwen, Mistral, and DeepSeek Method

ABSTRACT (IN ENGLISH)
Internal quality assurance systems are critical for the success of institutional accreditation in higher education. The current system still used manual assessment by crosschecking the qualitative and quantitative indicators. The assement time is more than a day, because there are 60 indicators in the document. This study develops an automated quality assessment specifically designed for the Technical Bachelor Degree to streamline the internal quality assurance process. The system integrates three Large Language Models (LLMs): Qwen, Mistral, and Deepseek to evaluate academic metrics against accreditation standards using seven samples of Tri Dharma indicators. The use of artificial intelligence in this study can overcome the inefficiencies of manual data management by replacing it with an AI-based analysis program specifically for internal academic quality assessment. The results show that Qwen achieves the highest precision with an MAE of 0.5714 and an RMSE of 0.7559. Mistral follows with an MAE of 0.7143 and an RMSE of 1.2536, while Deepseek demonstrates the fastest processing speed despite a higher deviation, recording an MAE of 1.81 and an RMSE of 2.153.

Keywords: *Instutional Accreditation; Tri Dharma; Automatic Quality Assessment; Large Language Models*

INTRODUCTION
The implementation of higher education in Indonesia is currently facing increasingly stringent quality standardization demands, where accreditation serves as the primary indicator of institutional credibilityThe Internal Quality Assurance System (SPMI) plays a vital role in ensuring that every study program meets the National Standards for Higher Education (SN-Dikti) and the criteria set by accreditation bodies such as BAN-PT or LAM (Efendi, Hapsari, & Yusup, 2025). However, digital transformation in quality assurance is not merely about digitizing documents, it represents a fundamental shift in how institutions manage data and overall organizational performance. The effectiveness of SPMI highly depends on the institution's ability to link accreditation processes with organizational performance through the utilization of integrated information technology (Sakuroh, Marlina, Krismayanti, & Adiningsih, 2026; Aljarallah & Dutta, 2022). Previous research was also conducted to assess higher education acreditation using different method, i.e., K-Means and OLAP methods (Sakinah, Syaputra, Rahman, Fajri, & Rachmansyah, 2025).

Technical study programs require rigorous and intensive quality monitoring to maintain their competitive edge (Subijanto, et al., 2021). Strengthening the quality assurance system is crucial to ensure that the aspects of Tri Dharma encompassing education, research, and community service maintain high quality standards regardless of the delivery method (Sugiarti, 2022). Without a consistent monitoring system, academic programs risk a decline in quality markers and face significant difficulties in fulfilling the physical evidence required during accreditation visitations (Ticolau, 2026).

The Primary obstacle frequently encountered is the reliance on manual assessment processes, which triggers "data silos" (fragmented data) and delayed reporting (Efendi, Hapsari, & Yusup, 2025). Recent studies indicate that conventional internal audit processes are often unable to capture the dynamic of large and fast paced academic data, making it difficult for study programs to make timely improvements (Isnaeni, Putranto, Andriyani, & Khomsah, 2025). Therefore, a comprehensive and automated data architecture is needed to directly process academic metrics for accreditation needs (Ticolau, 2026; Başaran, 2026). The integration of automated data analysis technology is expected to minimize human error and provide higher data transparency within the institutional framework (Sakuroh, Marlina, Krismayanti, & Adiningsih, 2026).

The advancement of Artificial Intelligence (AI) technology, particularly Large Language Models (LLMs), opens new opportunities for automated processing of complex academic data (Naveed, Khan, & Qui, 2025; Demir & Yavuz, 2026). In this study, three distinct AI models Qwen, Mistral, and DeepSeek exhibit varying performance

profiles when tasked with evaluating Tri Dharma metrics. While Qwen demonstrates high precision in error minimization, DeepSeek offers superior computational efficiency but with the fastest processing runtime (Aydin, Karaaslan, Erenay, & Bacanin, 2026; Jegham, Abdelatti, & Hendawi, 2025). In contrast, Mistral provides a balanced analytical approach but experiences higher latency during data processing (Nurohim, Setyadi, & Fauzi, 2025). These performance variances indicate that each model possesses unique strengths in terms of accuracy, speed, and consistency when handling data-based academic tasks.

According to this related work and problem, we conduct a comparative analysis of three AI engines to identify the most effective model for automated internal academic quality assessment in the Technical Bachelor Degeree. The methods used are Qwen, Deepseek, and Mistral to assess seven samples of tri dharma indicators (Joshi, 2025).

METHODS

Approach and Scope

This study uses qualitative and quantitative experimental approaches with a benchmarking method against three open-weight LLMs. The scope is limited to seven core Tri Dharma indicators: (1) Program Learning Outcomes (PLO) alignment and review, (2) Lesson Plan (RPS), (3) Learning Process, (4) Research and Community Service integration, (5) Capstone Design, (6) Research, and (7) Community Service. The subset comprises 4 qualitative and 3 quantitative indicators.

Operational Variable Definitions

The primary variable is the model score (ordinal scale 0–4) per indicator. Supporting variables include justification (text), confidence level (range 0–1), and runtime (seconds) per indicator.

Location and Sample

The study was conducted on a Technical Bachelor Study Program. The Self-Evaluation Report and Study Program Performance Report documents are the post-audit version from the 2024 cycle.

Materials and Main Tools

The system is built with:

- Embedding model paraphrase-multilingual-MiniLM-L12-v2 (384-d).
- Vector database ChromaDB with HNSW cosine similarity.
- LangGraph as the agent state-machine framework.
- Ollama as the inference backend on a 24 GB VRAM GPU server accessed via a Cloudflare-tunneSelf-Evaluation Report HTTPS endpoint.

The technical components described above are integrated to transform the Manual Flow (Baseline) into the Automated Flow (Proposed) as illustrated in Figure 1. The synergy between LangGraph and ChromaDB enables the system to transition seamlessly from data ingestion (Self-Evaluation Report PDF and Study Program Performance Report Excel) to RAG-based automated scoring. Supported by the Ollama inference backend, the system processes 60+ indicators simultaneously to generate justifications and confidence scores, which are then streamed into a CSV format for final validation by the assessor through a human-in-the-loop mechanism.

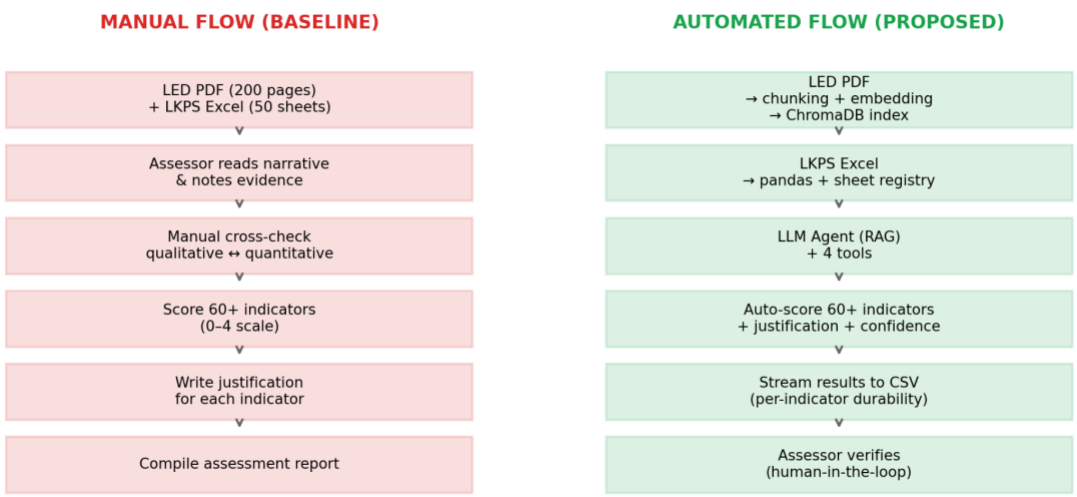


Figure 1. Comparison of Manual vs Automated Workflow

Data Collection and Analysis Techniques

Each indicator is evaluated once per model with identical sampling parameters, producing 21 invocations total (3 models × 7 indicators). Results are recorded directly to a CSV file. Reference scores are generated separately by ground-truth data. Analysis uses two metrics:

- Mean Absolute Error (MAE): $\frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$ (lower is better)
- Root Mean Squared Error (RMSE): $\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$ (lower is better)

RESULTS AND DISCUSSION

Model Performance Evaluation Results

This research conducts an automatic internal academic quality assessment of seven indicators of the Tri Dharma of Technical Bachelor Study Program. The assessment is carried out using three Large Language Model (LLM) models, namely Qwen, DeepSeek, and Mistral, with a score scale of 0 to 4. The documents being evaluated are the Study Program Performance Report and the Self-Evaluation Report. The performance test results of the three models are presented in Table 1.

Table 1. Performance of the LLM Model in Internal Academic Quality Assessment

Model	n	MAE	RMSE	AVG Confidence	AVG Runtime (Second)	N Errors
Qwen	7	0.5714	0.7559	0.7786	30.17	0
Mistral	7	0.7143	1.2536	0.7429	68.24	0
DeepSeek	7	1.81	2.153	0.8786	14.1	1

Based on Table 1, out of the three models tested against seven indicators (n = 7), the Qwen model shows the best performance in terms of assessment accuracy with the smallest Mean Absolute Error (MAE) value of 0.5714 and a Root Mean Square Error (RMSE) of 0.7559. The low MAE and RMSE values indicate that the scores produced by Qwen are closest to the reference scores, making this model the most reliable in the context of this quality assessment. The Mistral model ranks second with an MAE of 0.7143 and an RMSE of 1.2536, while DeepSeek has the highest MAE of 1.81 with an RMSE of 2.153, and it also produced one error (N Errors = 1) during the assessment process, which occurred on the Learning Process Plan indicator.

From the perspective of model confidence level, DeepSeek has the highest average confidence of 0.8786, followed by Mistral (0.8214) and Qwen (0.7786). Interestingly, the high confidence level in DeepSeek does not correlate with its score accuracy, which is actually the lowest among the three models. This indicates the phenomenon of overconfidence in the DeepSeek model, where the model assigns scores with high confidence even though the results deviate significantly from the reference score. This phenomenon is relevant to findings in several studies on LLM calibration, which show that the confidence score does not always represent the accuracy of predictions.

From the perspective of computational efficiency, DeepSeek has the fastest average runtime (14.1 seconds per indicator), followed by Qwen (30.17 seconds), and Mistral, which is the slowest with an average of 68.24 seconds per indicator. Thus, there is a trade-off between speed and accuracy: DeepSeek is fast but less accurate, while Qwen offers a better balance between speed and assessment accuracy.

Analysis of Indicator Scoring per Model

To obtain a more detailed Self-Evaluation Report picture, the results of the assessment of each model against the seven indicators of Tri Dharma are presented in Table 2.

Table 2. Qwen Model Scoring Indicators

No	Indicator	Score	Confidence Level	Run Time (seconds)
1	Relevance and Review of Program Learning Outcomes	3	0.75	23.5
2	Learning Process Plan	3	0.75	41.5
3	Learning Process	4	0.90	30.4
4	Integration of Research and Community Service in Learning	3	0.75	23.1
5	Field Program Signature Engineering Project (Capstone Design)	3	0.60	18.6
6	Research	4	0.75	19.9
7	Community Service	4	0.85	42.8
	Average	-	0.764	28.54

Based on Table 2, the Qwen model achieved the three highest scores (score 4) on the indicators of Learning Process, Research, and Community Service, while the other four indicators received a score of 3. The average confidence level of Qwen is 0.764 with an average runtime of 28.54 seconds per indicator. This model did not produce a single error during the assessment process, indicating stability in consistently processing all indicators. The lowest confidence level is found in the Capstone Design indicator (0.60), reflecting the complexity of assessing that indicator, which is technical and multi-component. Overall, Qwen's assessment is considered conservative yet consistent, with structured justifications based on documented evidence.

Table 3. DeepSeek Model Scoring Indicators

No	Indicator	Score	Confidence Level	Run Time (seconds)
1	Relevance and Review of Program Learning Outcomes	3	0.80	112.3
2	Learning Process Plan	3	0.50	18.4
3	Learning Process	3	0.80	64.9
4	Integration of Research and Community Service in Learning	1	0.80	11.6
5	Field Program Signature Engineering Project (Capstone Design)	4	1.00	12.3
6	Research	0	1.00	42.5
7	Community Service	0	0.80	55.1
	Average	-	0.814	45.3

Based on Table 3, the DeepSeek model shows the most unstable assessment pattern among the three models. Although it successfully scored 4 on the Capstone Design indicator, DeepSeek scored 0 on the Research and Community Service indicators, which deviates significantly from the reference value. This inaccuracy is further reinforced by the occurrence of one error on the Learning Process Plan indicator. DeepSeek's average confidence level is 0.814 with an average runtime of 45.3 seconds, slower than Qwen but faster than Mistral. The score of 0 given on the Research and Community Service indicators is likely due to the model's failure to recognize or interpret the context of relevant Self-Evaluation Report documents, a qualitative weakness that requires further prompt adjustments.

Table 4. Mistral Model Scoring Indicators

No	Indicator	Score	Confidence Level	Run Time (seconds)
1	Relevance and Review of Program Learning Outcomes	3	0.90	17.5
2	Learning Process Plan	2.33	0.85	15.3
3	Learning Process	0	0.90	17.9
4	Integration of Research and Community Service in Learning	0	0.80	13.4
5	Field Program Signature Engineering Project (Capstone Design)	4	0.50	8.5
6	Research	3	0.90	10.0
7	Community Service	0	0.90	11.9

	Average	-	0.821	13.5

Based on Table 4, the Mistral model has the most inconsistent performance in terms of accuracy. Three out of seven indicators received a score of 0, namely the Learning Process, Integration of Research and Community Service in Learning, and Community Service. This caused Mistral's MAE value to be the highest (0.7143 overall). Nevertheless, Mistral's average confidence level is 0.821, which paradoxically is the highest among the three for qualitative scoring, indicating that Mistral is quite confident in its answers despite many being inaccurate. Mistral's runtime is the fastest on average per indicator (13.5 seconds), but this speed is not matched by adequate assessment quality. One of the justifications produced by Mistral on the Capstone Design indicator is even contextually irrelevant, indicating that this model is prone to hallucination in the assessment of descriptive-qualitative indicators.

Break Down Analysis by Scoring Type

To gain a deeper understanding of the error patterns of each model, an analysis was conducted based on the type of indicators, namely qualitative indicators (4 indicators) and quantitative indicators (3 indicators). The breakdown results are presented in Table 5 and Figure 2.

Table 5. Breakdown by Scoring Type

Model	Tipe Skor	n	MAE	Score Zero Count
Qwen	Qualitative	4	0.75	0
Qwen	Quantitative	3	0.3333	0
DeepSeek	Qualitative	4	1.25	1
DeepSeek	Quantitative	3	0	0
Mistral	Qualitative	4	2.25	3
Mistral	Quantitative	3	1.2233	1

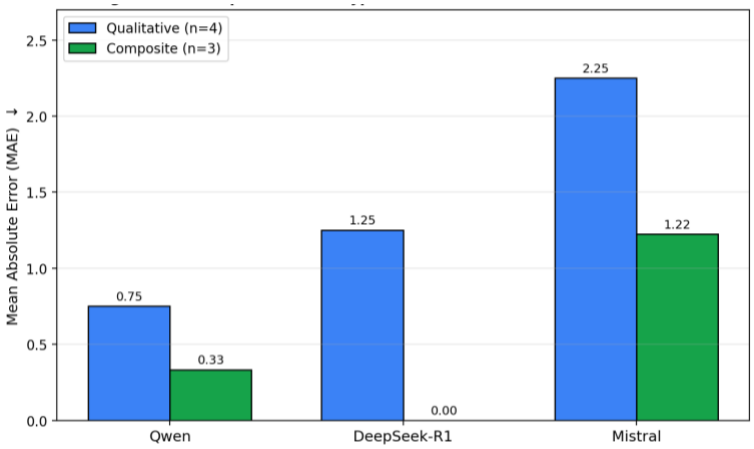


Figure 2. Comparison of MAE per Indicator Type

Based on Table 5 and Figure 2, the three models show a similar pattern, namely being superior in

handling quantitative indicators compared to qualitative ones. The Qwen model has a qualitative MAE of 0.75 and a quantitative MAE of 0.3333 with no zero scores in either. DeepSeek has a higher qualitative MAE (1.25) compared to its quantitative MAE (0), but it produced one zero score on the qualitative indicator. Mistral has the worst qualitative MAE (2.25) with three zero scores, and a quantitative MAE of 1.2233 with one zero score.

These findings indicate that the three LLMs tested consistently handle number and formula-based assessments (quantitative) better than context understanding, document interpretation, and logical reasoning-based assessments (qualitative). This aligns with the general characteristics of LLMs, which tend to be more accurate in tasks with explicit and structured rules. These findings also imply that the development of LLM-based quality assessment systems needs to pay special attention to the design of prompts and contexts for qualitative indicators.

Overall, the results of this study indicate that the Qwen model is the best choice for internal academic quality assessment based on LLM in the context of study program accreditation, with the lowest MAE value of 0.5714 and RMSE of 0.7559. This is in line with previous research that shows the Qwen model has a better ability to understand the context of language and academic documents compared to peer models.

The main weakness of DeepSeek in this study is its inability to accurately identify evidence from Study Program Performance Report and Self-Evaluation Report documents, particularly on indicators that require the synthesis of information across different parts of the document. These findings reveal a GAP between the retrieval and reasoning capabilities of LLMs in the domain of Technical Bachelor Study Program accreditation, which has not been extensively explored in the existing literature.

Meanwhile, Mistral shows weaknesses in terms of hallucination, where the model generates justifications that are irrelevant to the context of the assessed indicators. The phenomenon of hallucination in LLMs for domain-specific tasks has been identified as a major challenge that needs to be addressed through techniques such as Retrieval-Augmented Generation (RAG) or domain-based data fine-tuning.

Theoretically, this research contributes to proving that the automation of internal academic quality assessment using LLM is a viable approach, although it still requires further refinement, especially on qualitative indicators. Implementationally, this system has the potential to accelerate the accreditation process and reduce the administrative burden on quality assurance teams at Technical Bachelor Study Program. This research delivers four complementary novelty contributions that distinguish it from the existing LLM-Powered Automated Assessment literature (Sun et al., 2025; MDPI 2025):

1. **Domain novelty:** the first study applying open-weight LLM agents to Indonesian LAM-style higher-education accreditation, whereas prior work is dominated by English-language student-work grading (essays, code) (arXiv 2508.02442).
2. **Cross-modal RAG novelty:** an agent architecture that cross-references qualitative documents (Self-Evaluation Report narrative PDF) with quantitative documents (Study Program Performance Report structured Excel) through four specialized tools, while the 2025 RAG-in-education survey covers only single-modal text-to-text systems (Gao et al., 2025).
3. **Tool-calling paradigm comparison:** a side-by-side comparison of native function calling versus prompt-based JSON dispatch on the same domain-specific task, filling a gap not addressed by the generic BFCL v4 or ToolBench evaluations.
4. **Calibration finding:** empirical documentation of overconfidence in the DeepSeek-R1

reasoning model on institutional document scoring, extending recent calibration literature (HAC, HalluScan, T3) to a new domain.

CONCLUSION

This research aims to develop an automatic internal academic quality assessment system using three Large Language Model (LLM) models, namely Qwen, DeepSeek, and Mistral, in order to support the accreditation process of Technical Bachelor Study Program. The assessment is conducted on seven indicators of the Tri Dharma of Higher Education based on the Study Program Performance Report and Self-Evaluation Report documents with a score scale of 0–4. Based on the research results and discussions that have been outlined, the following conclusions can be drawn :

- The Qwen model proved to be the most accurate model in assessing internal academic quality, with the smallest MAE value of 0.5714 and an RMSE of 0.7559. Qwen did not produce any errors during the assessment process, demonstrating high stability and consistency across all Tri Dharma indicators.
- The DeepSeek model has the highest computational speed with an average runtime of 14.1 seconds per indicator and the highest average confidence level (0.8786), but its accuracy is the lowest with an MAE of 1.81 and an RMSE of 2.153. There is a phenomenon of overconfidence, where the model gives scores with high confidence despite deviating significantly from the reference value, accompanied by one error in the Learning Process Plan indicator.
- The Mistral model shows the greatest instability, with three out of seven indicators scoring 0 (Learning Process, Research and Community Service Integration, and Community Service), and indications of hallucination found in the justification of the Capstone Design indicator. Nevertheless, Mistral has the fastest average runtime per indicator (13.5 seconds).
- The three models consistently show better performance on quantitative indicators compared to qualitative ones. This indicates that current LLMs are more reliable in evaluating criteria based on numbers and formulas compared to criteria that require contextual interpretation and in-depth descriptive reasoning.
- Overall, the order of model accuracy from best to worst is Qwen, followed by Mistral, and DeepSeek. This LLM-based assessment automation system has proven to be feasible as an alternative approach to expedite and lighten the accreditation process, particularly for distance education program providers.

Theoretically, this research contributes to proving that the automation of internal academic quality assessment using LLM is a viable approach, although it still requires further refinement, especially on qualitative indicators. In terms of implementation, this system has the potential to accelerate the accreditation process and reduce the administrative burden on the quality assurance team at universities. For future research, it is recommended to further explore Retrieval-Augmented Generation techniques and domain-specific fine-tuning to improve accuracy on qualitative indicators, as well as to expand the scope of study programs and types of accreditation documents evaluated.

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